Final Project

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Database Management Systems

1) Introduction to the ArtBase Database Project

- ArtBase is a specialized database solution tailored for art galleries, facilitating efficient management of collections and customer relationships
- Art galleries often encounter logistical challenges in cataloging artworks, understanding customer preferences, and maintaining operational efficiency
- ArtBase aims to address these challenges by offering tools to organize artist information, track artwork details, categorize pieces, and maintain customer profiles.
- This project focuses on conceptualizing, designing, and implementing the ArtBase database system to meet the unique needs of art galleries, emphasizing entity-relationship modeling, relational schema design, SQL query development, and optional application development.

2) Conceptual Design

In the conceptual design phase of the ArtBase Database Project, we identify the entity types and relationship types crucial for managing the provided dataset effectively. This phase establishes the foundational structure of the database schema.

Entity Types:

1) Artist
    - Artist Names (Unique identifier)
    - Birthplaces
    - Age
    - Style of Art
2) Artwork
    - Title of Artwork (Unique identifier)
    - Year
    - Type of Art
    - Price
    - Group Id (Foreign key referencing Art Group)
3) Art Group
    - Group Id (Unique identifier)
    - Group Name
4) Customer

- Customer Name (Unique identifier)
- Address
- Total Amount Spent

Relationship Types:

1) Artist creates Artwork
- One-to-Many relationship: An artist can create multiple artworks.
2) Artwork belongs to Art Group
- Many-to-One relationship: Multiple artworks can belong to the same art group.
3) Customer purchases Artwork
- One-to-Many relationship: A customer can purchase multiple artworks.
4) Customer has preferences for Artists and Art Groups
- Many-to-Many relationship: A customer can have preferences for multiple artists and art groups, and an artist/art group can be preferred by multiple customers.

Data Dictionary:

Artist:

- Artist Names
- Birthplaces
- Age
- Style of Art

Artwork:

- Title of Artwork
- Year
- Type of Art
- Price
- Group Id

Art Group:

- Group Id
- Group Name

Customer:

- Customer Name
- Address
- Total Amount Spent

Data Constraints and Assumptions:

Each artist name and artwork title is unique.

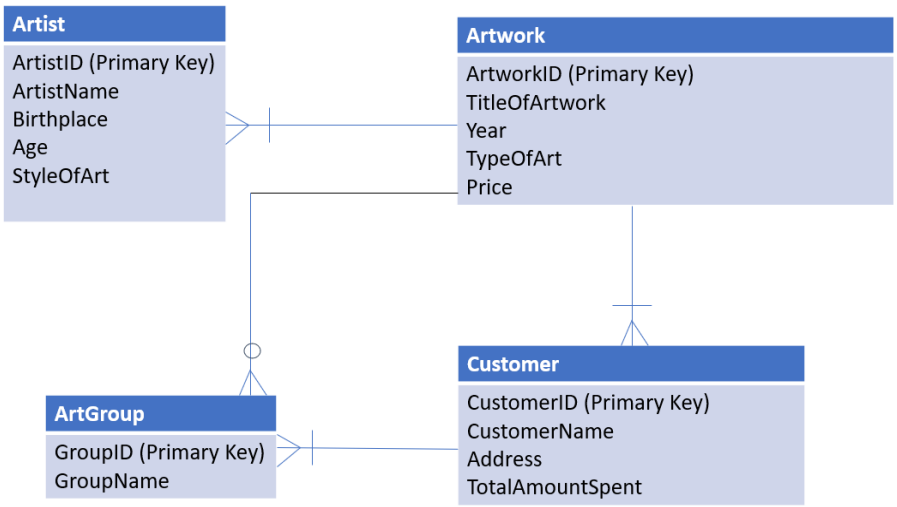
Customers may have preferences for multiple artists and art groups.

Total amount spent by a customer is cumulative across all transactions.

This conceptual model provides a clear representation of the entities and relationships within the dataset, laying the groundwork for the subsequent design phases of the ArtBase Database Project.

3) Logical design –

ER Diagram:



Relational Schema:

1) Artist
- ArtistID (Primary Key)
- ArtistName
- Birthplace
- Age
- StyleOfArt

2) Artwork
- ArtworkID (Primary Key)
- TitleOfArtwork
- Year
- TypeOfArt
- Price
- GroupID (Foreign Key referencing ArtGroup)

3) ArtGroup
- GroupID (Primary Key)
- GroupName

4) Customer

- CustomerID (Primary Key)
- CustomerName
- Address
- TotalAmountSpent

Key Definitions:

- Primary Key (PK): Unique identifier for each record in the table.
- Foreign Key (FK): Column(s) in a table that reference the primary key of another table. It establishes a link between tables.

Relationships:

a) Artist creates Artwork:
- ArtistID (in Artwork table) is a Foreign Key referencing ArtistID in the Artist table.

b) Artwork belongs to Art Group:
- GroupID (in Artwork table) is a Foreign Key referencing GroupID in the ArtGroup table.

Constraints:

1) Primary Keys:
- ArtistID (Artist table)
- ArtworkID (Artwork table)
- GroupID (ArtGroup table)
- CustomerID (Customer table)

2) Foreign Keys:
- GroupID (Artwork table referencing ArtGroup table)
- ArtistID (Artwork table referencing Artist table)

This logical design provides a structured representation of the database schema, with clearly defined tables, keys, and relationships. It serves as a crucial step in the development of the ArtBase Database, ensuring data integrity and efficient data retrieval operations.

4) Physical database design – A description of all tables and attributes in your database. You have to normalize or de-normalize your data tables that make sense your case based on your and clients' requirements. Specify all functional dependencies, constrains, keys (primary and foreign) and normalization decision and issues for your database. (25 points)

Physical Database Design

In the physical database design phase of the ArtBase Database Project, we provide a detailed description of all tables and attributes, including normalization decisions, functional dependencies, constraints, and keys. This phase ensures that the database schema is optimized for efficient storage, retrieval, and maintenance of data.

Table Descriptions:

1. **Artist**
    - Attributes:
        - ArtistID (Primary Key)

        - ArtistName
        - Birthplace
        - Age
        - StyleOfArt
2. **Artwork**
    - Attributes:
        - ArtworkID (Primary Key)
        - TitleOfArtwork
        - Year
        - TypeOfArt
        - Price
        - GroupID (Foreign Key)
3. **ArtGroup**
    - Attributes:
        - GroupID (Primary Key)
        - GroupName
4. **Customer**
    - Attributes:
        - CustomerID (Primary Key)
        - CustomerName
        - Address
        - TotalAmountSpent

Normalization Decisions:

- **1st Normal Form (1NF):** All attributes are atomic and do not contain repeating groups.
- **2nd Normal Form (2NF):** All non-key attributes are fully functionally dependent on the primary key.
- **3rd Normal Form (3NF):** No transitive dependencies exist; all non-key attributes depend only on the primary key.

Functional Dependencies:

- ArtistID → ArtistName, Birthplace, Age, StyleOfArt
- ArtworkID → TitleOfArtwork, Year, TypeOfArt, Price, GroupID

- GroupID → GroupName
- CustomerID → CustomerName, Address, TotalAmountSpent

Constraints:

- Primary Keys:
    - ArtistID (Artist)
    - ArtworkID (Artwork)
    - GroupID (ArtGroup)
    - CustomerID (Customer)
- Foreign Keys:
    - GroupID (Artwork referencing ArtGroup)
    - ArtistID (Artwork referencing Artist)

Normalization Decision:

- The tables are normalized up to the 3rd Normal Form to eliminate redundancy and maintain data integrity.
- Each table contains only the attributes necessary for its purpose, reducing data redundancy and improving efficiency.
- The use of foreign keys establishes relationships between tables, ensuring data consistency.

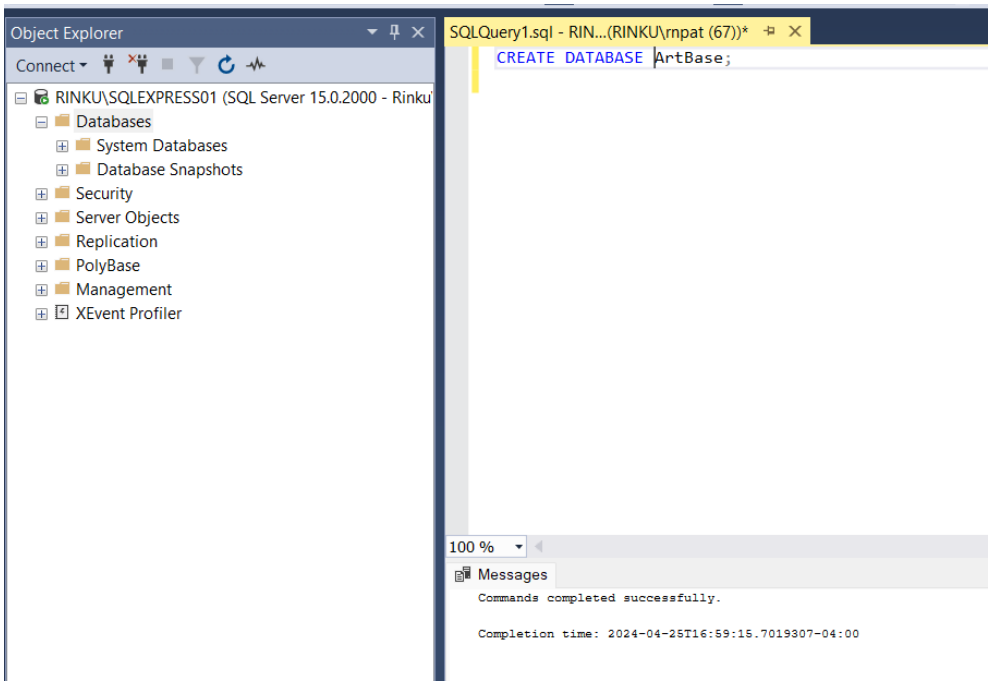
Issues:

- De-normalization: In certain cases where performance considerations outweigh normalization principles, de-normalization may be applied. However, in this case, de-normalization is not necessary as the normalized schema adequately meets the requirements and maintains data integrity.

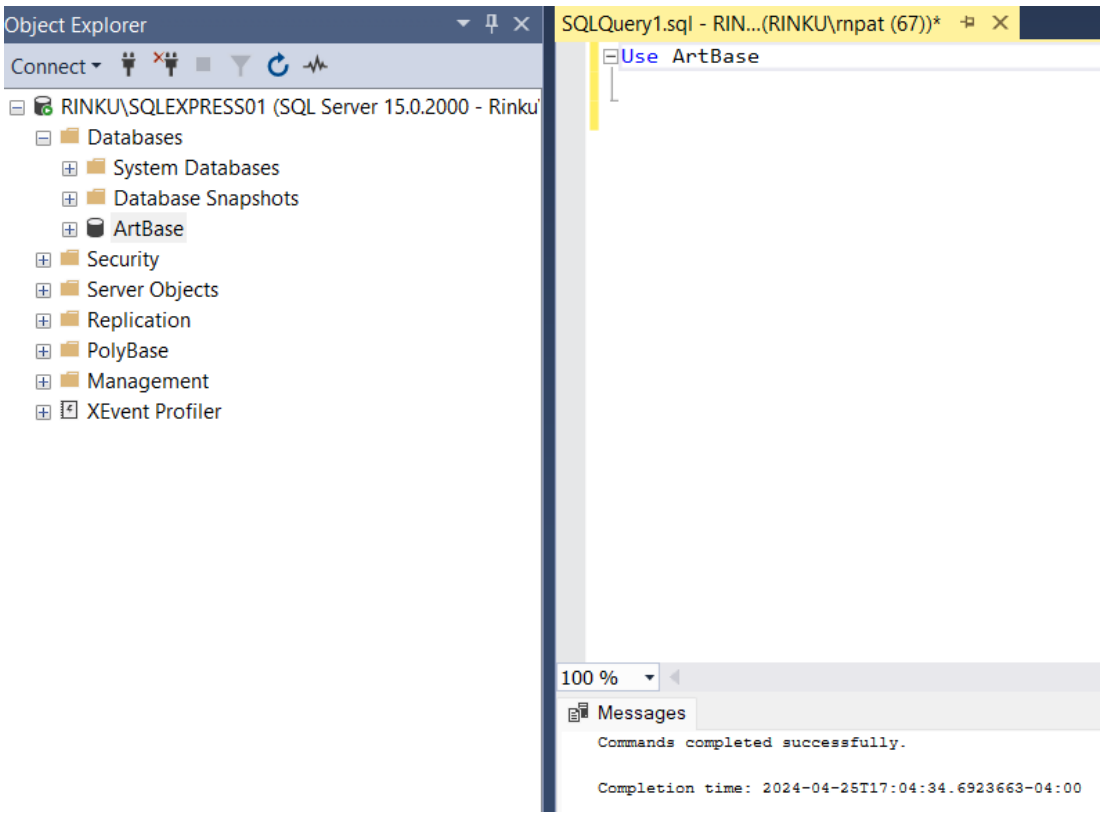
This physical database design ensures that the database schema is well-structured, normalized, and optimized for efficient data storage and retrieval. It forms the foundation for the implementation of the ArtBase Database, facilitating seamless operations for art galleries and their stakeholders.

5) SQL queries – You have to develop SQL queries that can perform Select, Insert, Update, Delete data records, and Join two or more tables. Stored procedures and triggers are highly recommended to be used. (25 points)

Create new database:

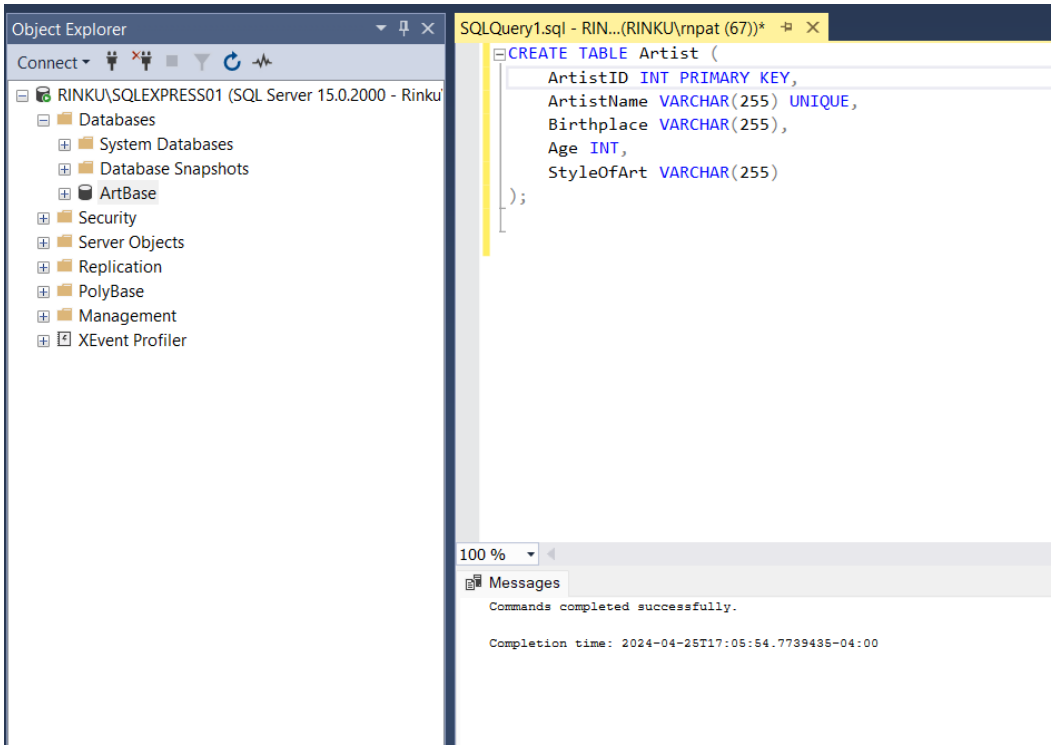


Use new Database: ArtBase



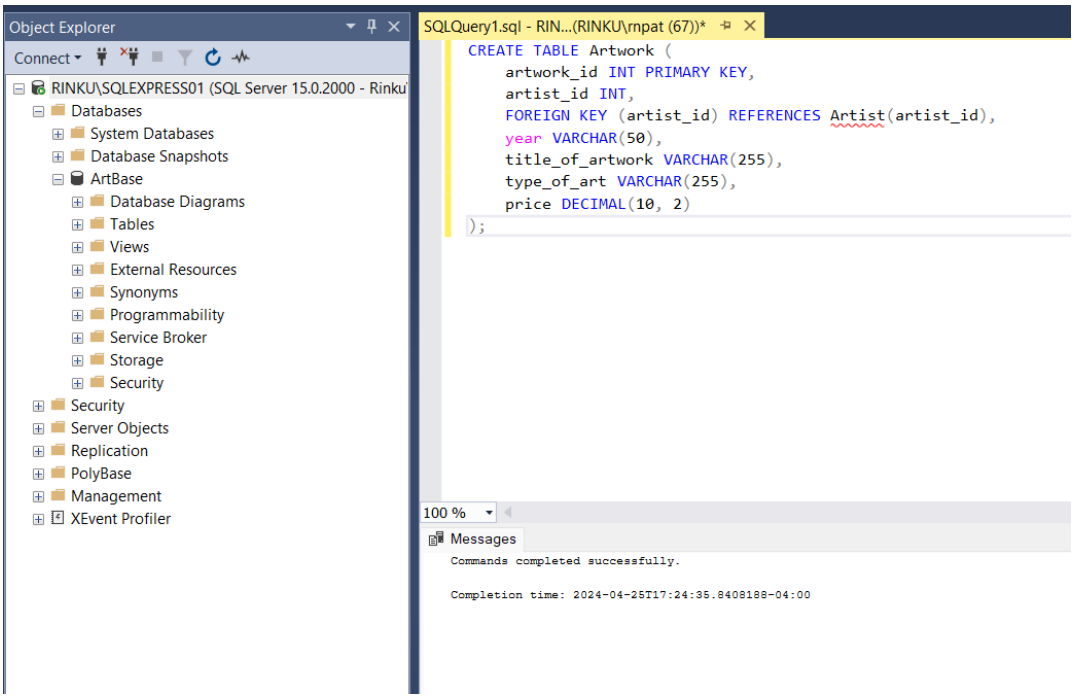
Create Artist table:

```
CREATE TABLE Artist (
    ArtistID INT PRIMARY KEY,
    ArtistName VARCHAR(255) UNIQUE,
    Birthplace VARCHAR(255),
    Age INT,
    StyleOfArt VARCHAR(255)
);
```



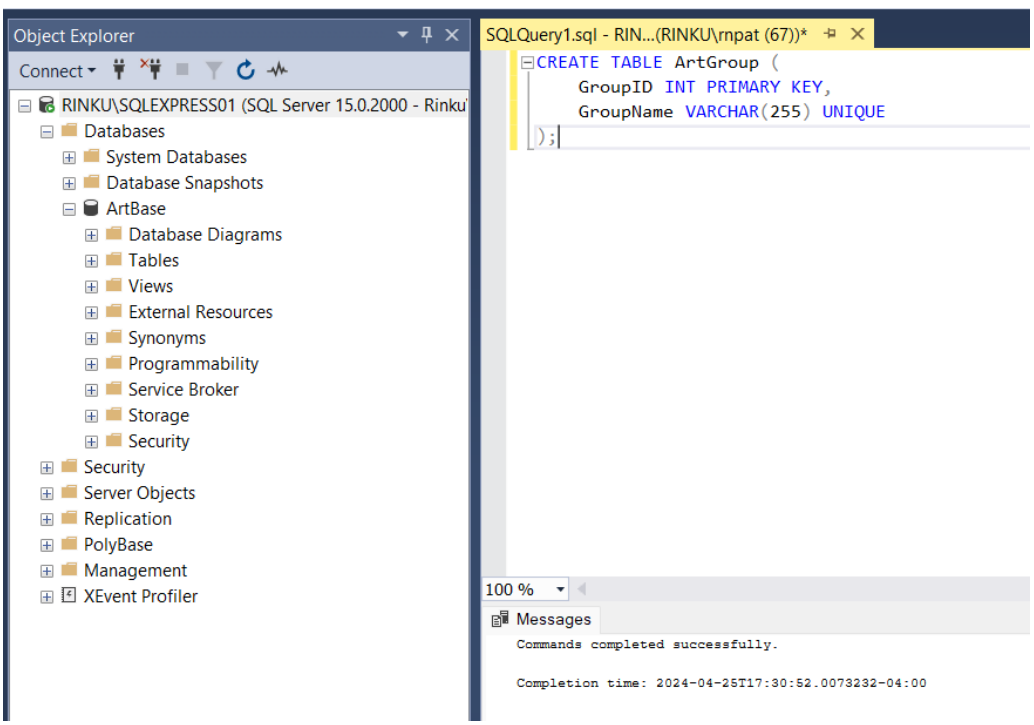
Create Artwork table:

```
CREATE TABLE Artwork (
    artwork_id INT PRIMARY KEY,
    artist_id INT,
    FOREIGN KEY (artist_id) REFERENCES Artist(artist_id),
    year VARCHAR(50),
    title_of_artwork VARCHAR(255),
    type_of_art VARCHAR(255),
    price DECIMAL(10, 2)
       );
```



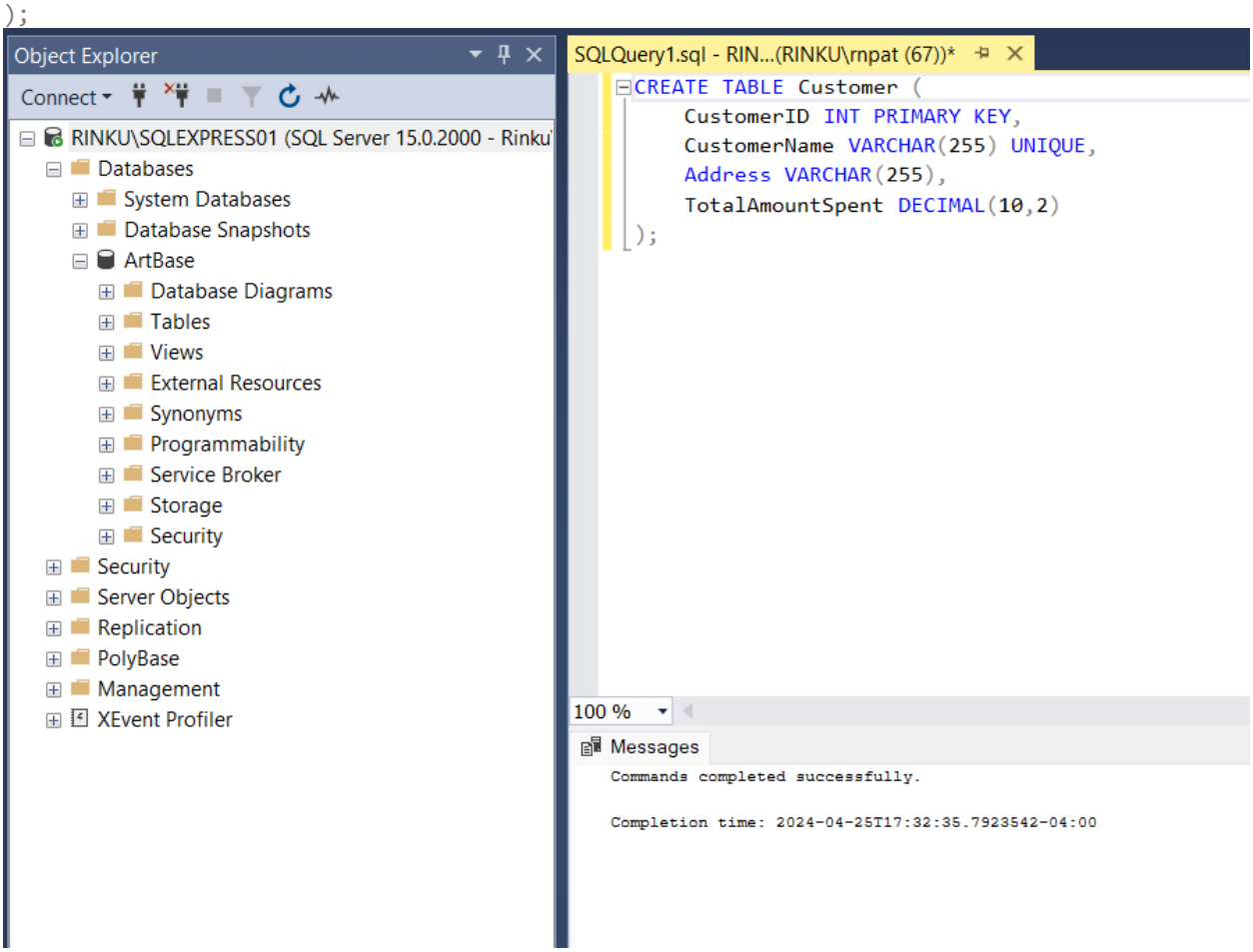
Create Group table:

```
CREATE TABLE ArtGroup (
    GroupID INT PRIMARY KEY,
    GroupName VARCHAR(255) UNIQUE
    );
```



Create Customer table:

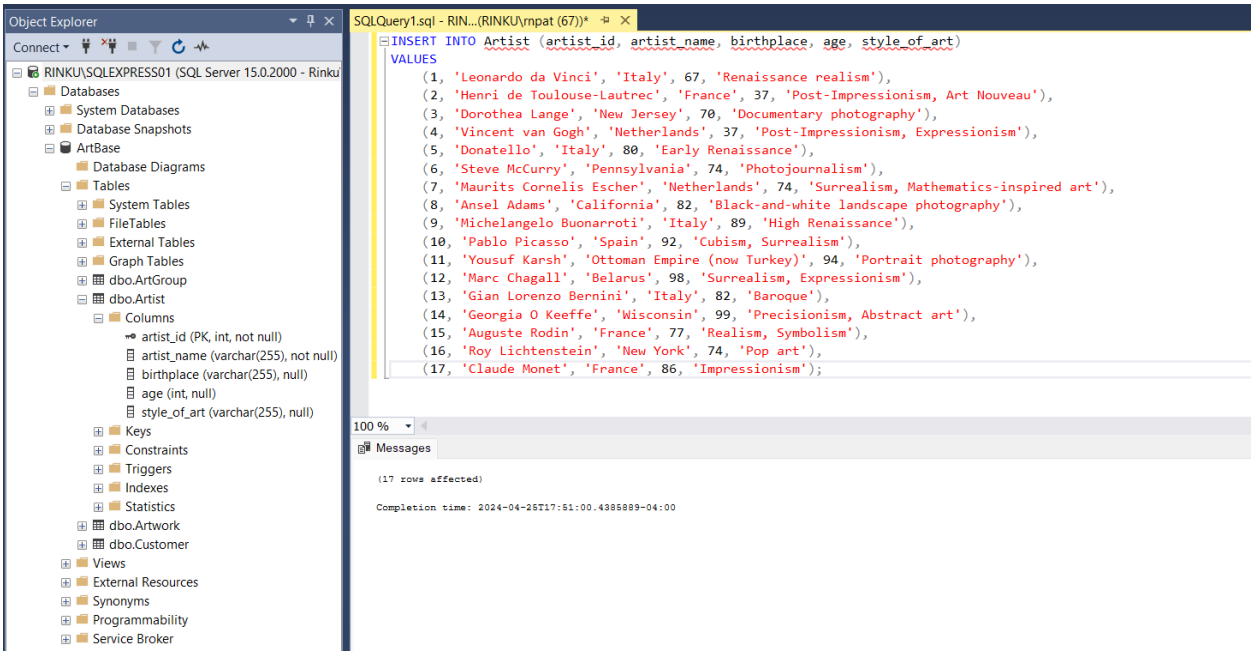
```
CREATE TABLE Customer (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(255) UNIQUE,
    Address VARCHAR(255),
    TotalAmountSpent DECIMAL(10,2)
    );
```



Insert data into Artist table:

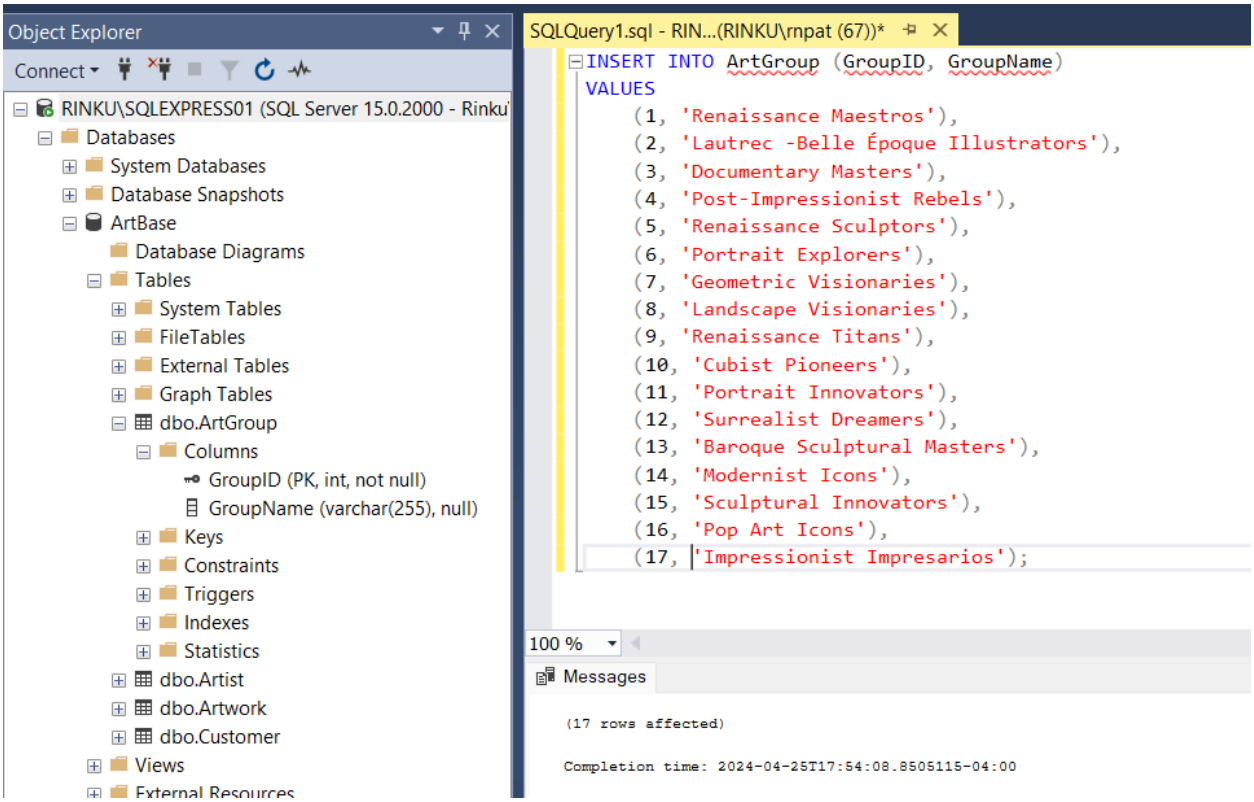
```
INSERT INTO Artist (artist_id, artist_name, birthplace, age, style_of_art)
VALUES
    (1, 'Leonardo da Vinci', 'Italy', 67, 'Renaissance realism'),
    (2, 'Henri de Toulouse-Lautrec', 'France', 37, 'Post-Impressionism, Art Nouveau'),
    (3, 'Dorothea Lange', 'New Jersey', 70, 'Documentary photography'),
    (4, 'Vincent van Gogh', 'Netherlands', 37, 'Post-Impressionism, Expressionism'),
    (5, 'Donatello', 'Italy', 80, 'Early Renaissance'),
    (6, 'Steve McCurry', 'Pennsylvania', 74, 'Photojournalism'),
    (7, 'Maurits Cornelis Escher', 'Netherlands', 74, 'Surrealism, Mathematics-inspired
art'),
    (8, 'Ansel Adams', 'California', 82, 'Black-and-white landscape photography'),
    (9, 'Michelangelo Buonarroti', 'Italy', 89, 'High Renaissance'),
    (10, 'Pablo Picasso', 'Spain', 92, 'Cubism, Surrealism'),
    (11, 'Yousuf Karsh', 'Ottoman Empire (now Turkey)', 94, 'Portrait photography'),
```

```
    (12, 'Marc Chagall', 'Belarus', 98, 'Surrealism, Expressionism'),
    (13, 'Gian Lorenzo Bernini', 'Italy', 82, 'Baroque'),
    (14, 'Georgia O Keeffe', 'Wisconsin', 99, 'Precisionism, Abstract art'),
    (15, 'Auguste Rodin', 'France', 77, 'Realism, Symbolism'),
    (16, 'Roy Lichtenstein', 'New York', 74, 'Pop art'),
          (17, 'Claude Monet', 'France', 86, 'Impressionism');
```

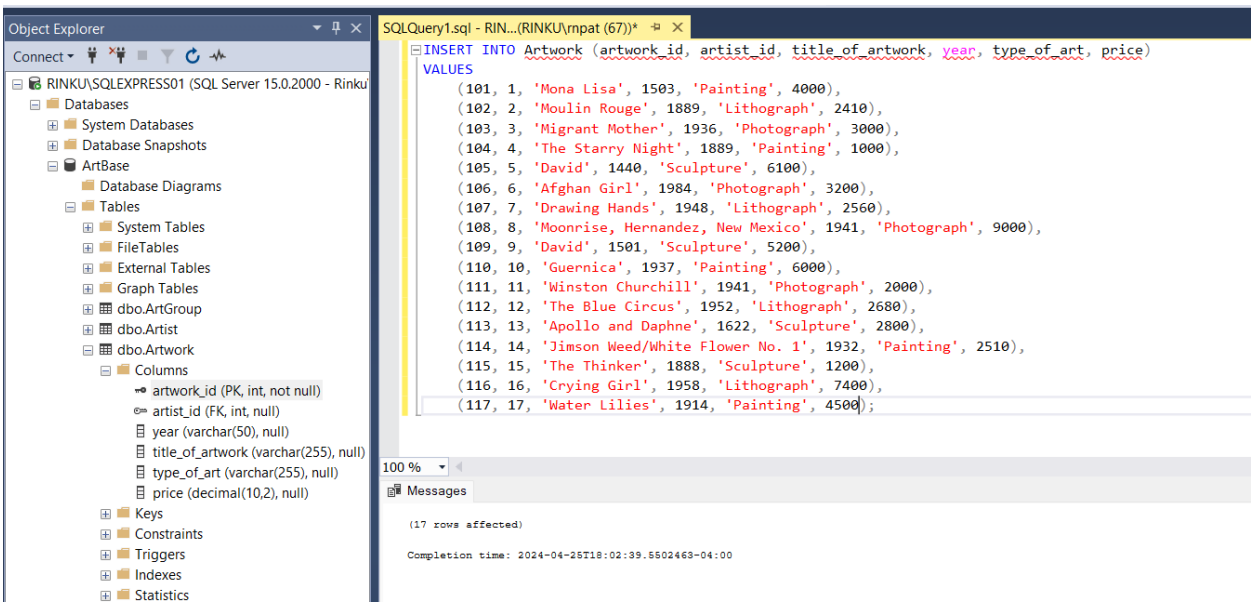


INSERT INTO ArtGroup (GroupName):

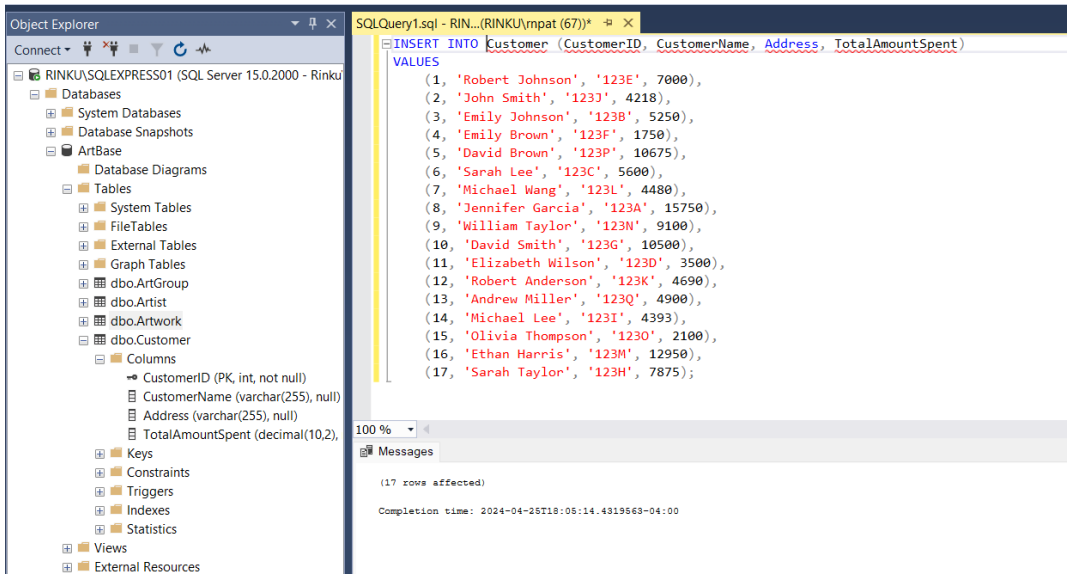
```
INSERT INTO ArtGroup (GroupID, GroupName)
VALUES
    (1, 'Renaissance Maestros'),
    (2, 'Lautrec -Belle Époque Illustrators'),
    (3, 'Documentary Masters'),
    (4, 'Post-Impressionist Rebels'),
    (5, 'Renaissance Sculptors'),
    (6, 'Portrait Explorers'),
    (7, 'Geometric Visionaries'),
    (8, 'Landscape Visionaries'),
    (9, 'Renaissance Titans'),
    (10, 'Cubist Pioneers'),
    (11, 'Portrait Innovators'),
    (12, 'Surrealist Dreamers'),
    (13, 'Baroque Sculptural Masters'),
    (14, 'Modernist Icons'),
    (15, 'Sculptural Innovators'),
    (16, 'Pop Art Icons'),
       (17, 'Impressionist Impresarios');
```




```sql
INSERT INTO Artwork (artwork_id, artist_id, title_of_artwork, year, type_of_art, price)
VALUES
    (101, 1, 'Mona Lisa', 1503, 'Painting', 4000),
    (102, 2, 'Moulin Rouge', 1889, 'Lithograph', 2410),
    (103, 3, 'Migrant Mother', 1936, 'Photograph', 3000),
    (104, 4, 'The Starry Night', 1889, 'Painting', 1000),
    (105, 5, 'David', 1440, 'Sculpture', 6100),
    (106, 6, 'Afghan Girl', 1984, 'Photograph', 3200),
    (107, 7, 'Drawing Hands', 1948, 'Lithograph', 2560),
    (108, 8, 'Moonrise, Hernandez, New Mexico', 1941, 'Photograph', 9000),
    (109, 9, 'David', 1501, 'Sculpture', 5200),
    (110, 10, 'Guernica', 1937, 'Painting', 6000),
    (111, 11, 'Winston Churchill', 1941, 'Photograph', 2000),
    (112, 12, 'The Blue Circus', 1952, 'Lithograph', 2680),
    (113, 13, 'Apollo and Daphne', 1622, 'Sculpture', 2800),
    (114, 14, 'Jimson Weed/White Flower No. 1', 1932, 'Painting', 2510),
    (115, 15, 'The Thinker', 1888, 'Sculpture', 1200),
    (116, 16, 'Crying Girl', 1958, 'Lithograph', 7400),
          (117, 17, 'Water Lilies', 1914, 'Painting', 4500);
```




```sql
INSERT INTO Customer (CustomerID, CustomerName, Address, TotalAmountSpent)
VALUES
    (1, 'Robert Johnson', '123E', 7000),
    (2, 'John Smith', '123J', 4218),
    (3, 'Emily Johnson', '123B', 5250),
    (4, 'Emily Brown', '123F', 1750),
    (5, 'David Brown', '123P', 10675),
    (6, 'Sarah Lee', '123C', 5600),
    (7, 'Michael Wang', '123L', 4480),
    (8, 'Jennifer Garcia', '123A', 15750),
    (9, 'William Taylor', '123N', 9100),
    (10, 'David Smith', '123G', 10500),
    (11, 'Elizabeth Wilson', '123D', 3500),
    (12, 'Robert Anderson', '123K', 4690),
    (13, 'Andrew Miller', '123Q', 4900),
    (14, 'Michael Lee', '123I', 4393),
    (15, 'Olivia Thompson', '123O', 2100),
    (16, 'Ethan Harris', '123M', 12950),
          (17, 'Sarah Taylor', '123H', 7875);
```

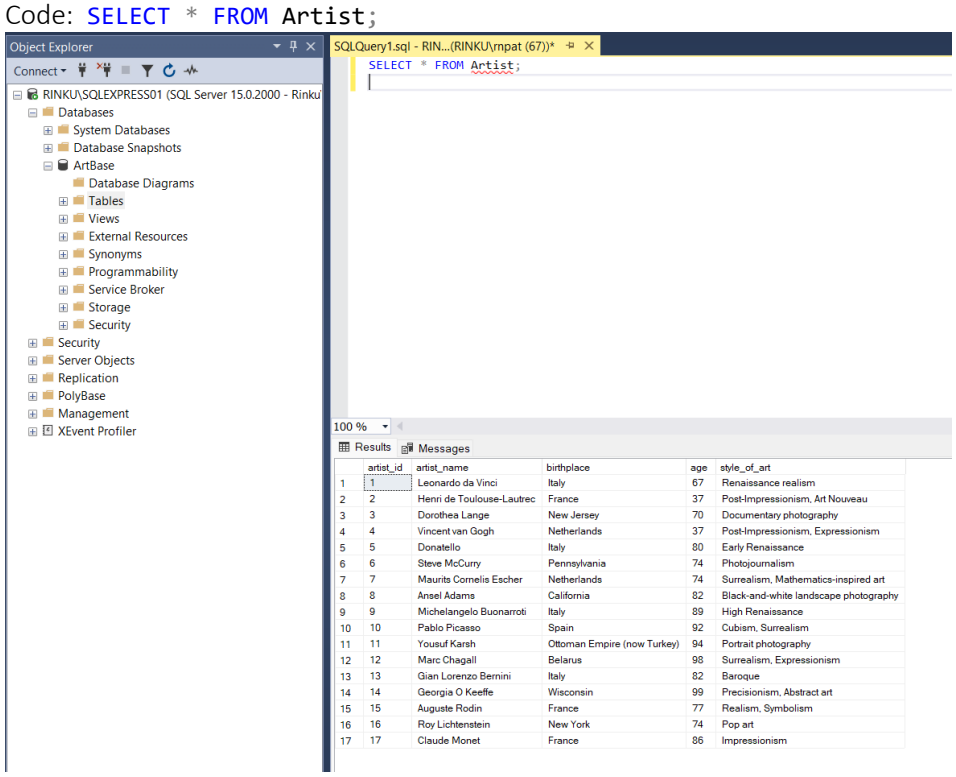


SQL Queries

In this section, we'll develop SQL queries to perform various operations on the ArtBase Database, including Select, Insert, Update, and Delete data records. Additionally, we'll demonstrate the use of Joins to retrieve data from multiple tables. Stored procedures and triggers will also be utilized for enhanced functionality and data integrity.

1. **Select Queries:**
   - Retrieve all artists:

Code: `SELECT * FROM Artist;`

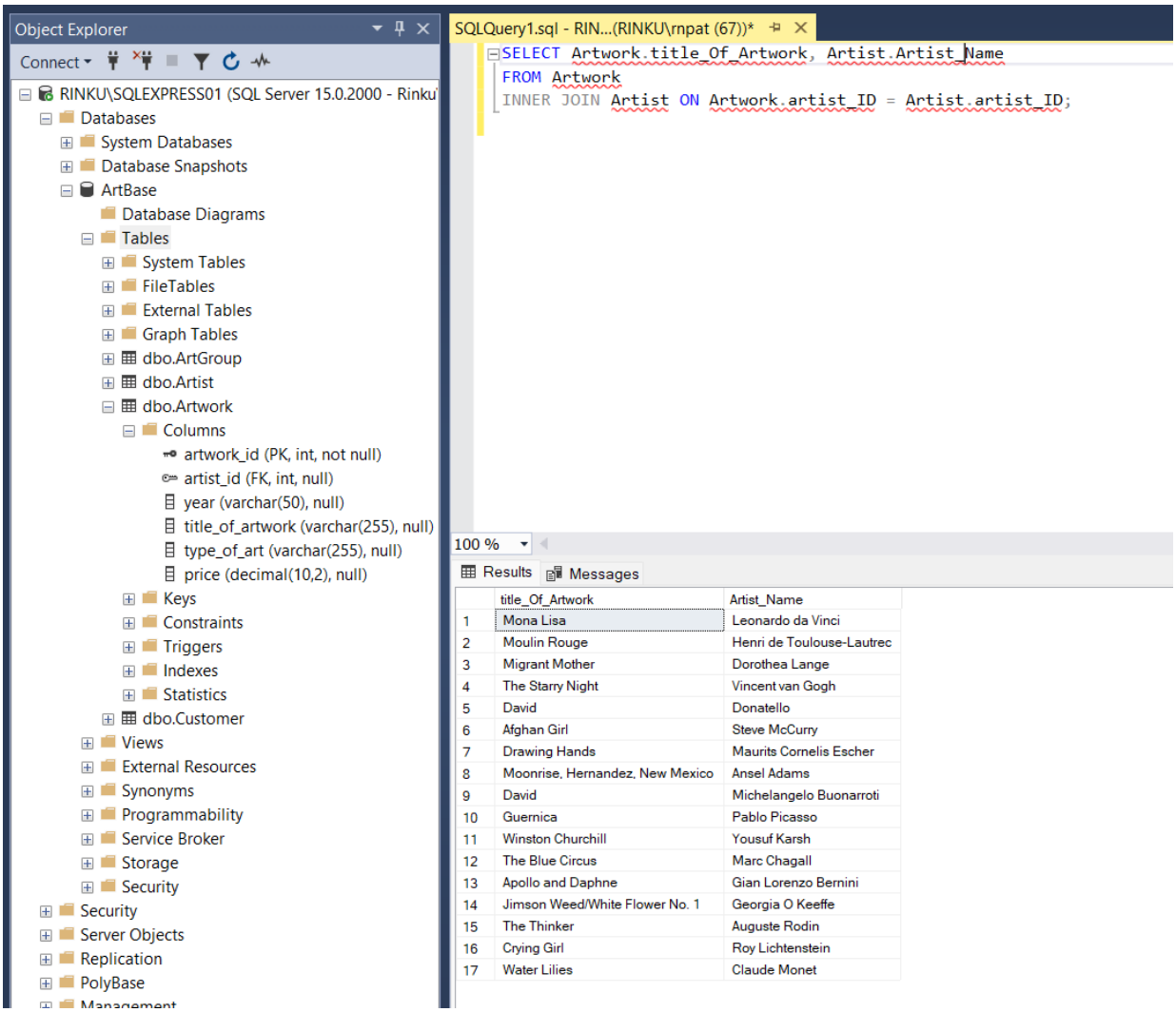


	artist_id	artist_name	birthplace	age	style_of_art
1	1	Leonardo da Vinci	Italy	67	Renaissance realism
2	2	Henri de Toulouse-Lautrec	France	37	Post-Impressionism, Art Nouveau
3	3	Dorothea Lange	New Jersey	70	Documentary photography
4	4	Vincent van Gogh	Netherlands	37	Post-Impressionism, Expressionism
5	5	Donatello	Italy	80	Early Renaissance
6	6	Steve McCurry	Pennsylvania	74	Photojournalism
7	7	Maurits Cornelis Escher	Netherlands	74	Surrealism, Mathematics-inspired art
8	8	Ansel Adams	California	82	Black-and-white landscape photography
9	9	Michelangelo Buonarroti	Italy	89	High Renaissance
10	10	Pablo Picasso	Spain	92	Cubism, Surrealism
11	11	Yousuf Karsh	Ottoman Empire (now Turkey)	94	Portrait photography
12	12	Marc Chagall	Belarus	98	Surrealism, Expressionism
13	13	Gian Lorenzo Bernini	Italy	82	Baroque
14	14	Georgia O Keeffe	Wisconsin	99	Precisionism, Abstract art
15	15	Auguste Rodin	France	77	Realism, Symbolism
16	16	Roy Lichtenstein	New York	74	Pop art
17	17	Claude Monet	France	86	Impressionism

- Retrieve artworks with their corresponding artist names:

Code:

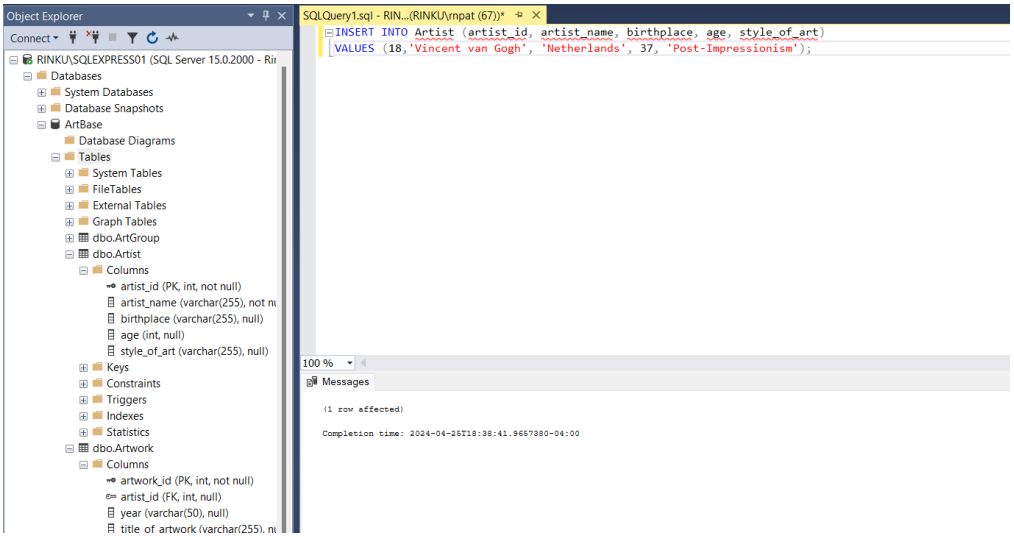
```
SELECT Artwork.title_Of_Artwork, Artist.Artist_Name
FROM Artwork
INNER JOIN Artist ON Artwork.artist_ID = Artist.artist_ID;
```

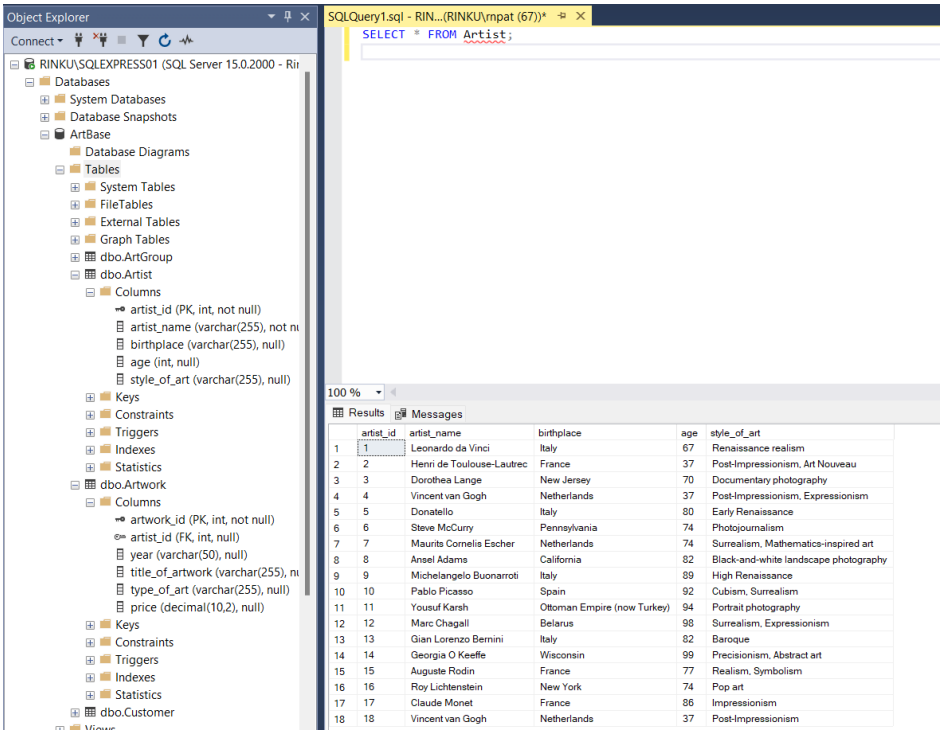


2. **Insert Queries:**
    - Insert a new artist:

Code:

```
INSERT INTO Artist (artist_id, artist_name, birthplace, age, style_of_art)
VALUES (18,'Vincent van Gogh', 'Netherlands', 37, 'Post-Impressionism');
```



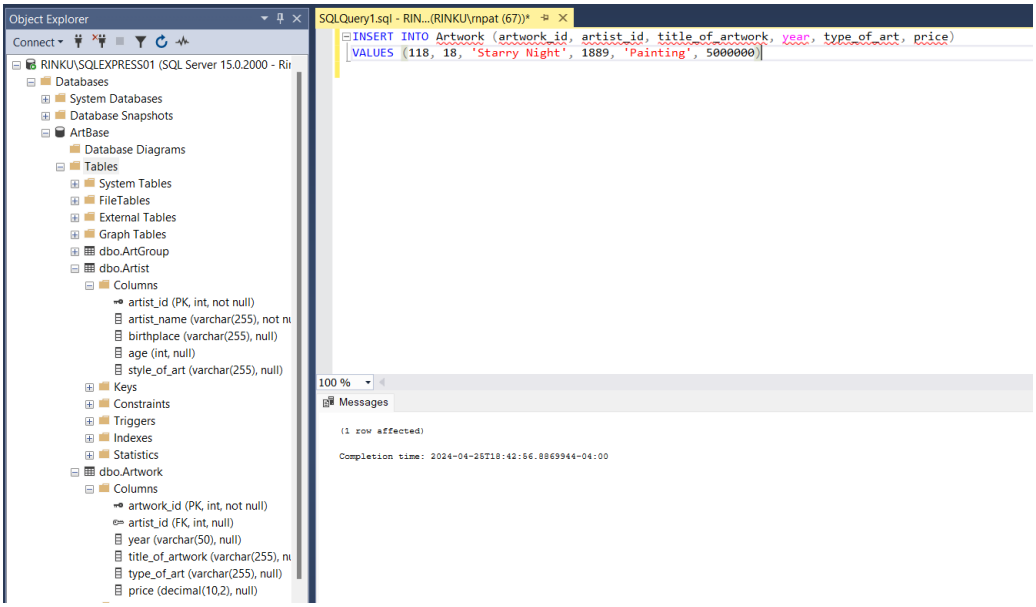


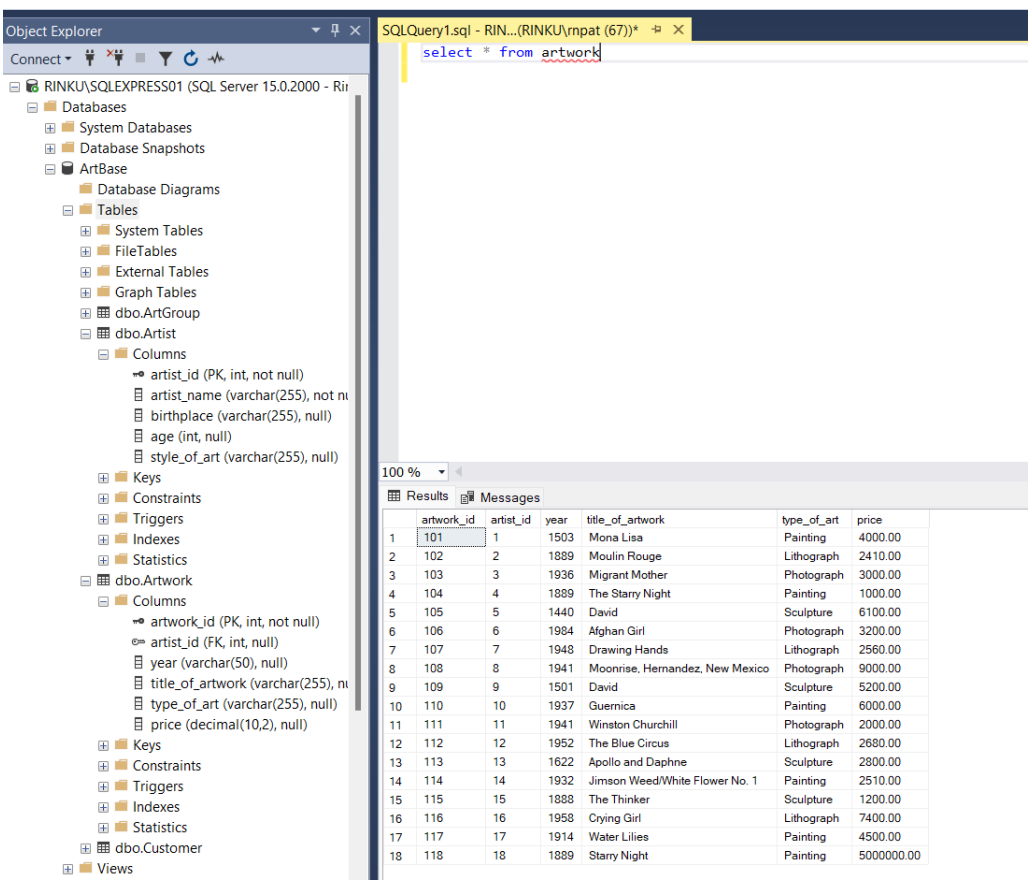
	artist_id	artist_name	birthplace	age	style_of_art
1	1	Leonardo da Vinci	Italy	67	Renaissance realism
2	2	Henri de Toulouse-Lautrec	France	37	Post-Impressionism, Art Nouveau
3	3	Dorothea Lange	New Jersey	70	Documentary photography
4	4	Vincent van Gogh	Netherlands	37	Post-Impressionism, Expressionism
5	5	Donatello	Italy	80	Early Renaissance
6	6	Steve McCurry	Pennsylvania	74	Photojournalism
7	7	Maurits Cornelis Escher	Netherlands	74	Surrealism, Mathematics-inspired art
8	8	Ansel Adams	California	82	Black-and-white landscape photography
9	9	Michelangelo Buonarroti	Italy	89	High Renaissance
10	10	Pablo Picasso	Spain	92	Cubism, Surrealism
11	11	Yousuf Karsh	Ottoman Empire (now Turkey)	94	Portrait photography
12	12	Marc Chagall	Belarus	98	Surrealism, Expressionism
13	13	Gian Lorenzo Bernini	Italy	82	Baroque
14	14	Georgia O Keeffe	Wisconsin	99	Precisionism, Abstract art
15	15	Auguste Rodin	France	77	Realism, Symbolism
16	16	Roy Lichtenstein	New York	74	Pop art
17	17	Claude Monet	France	86	Impressionism
18	18	Vincent van Gogh	Netherlands	37	Post-Impressionism

- Insert a new artwork:

Code:

```
INSERT INTO Artwork (artwork_id, artist_id, title_of_artwork, year, type_of_art, price)
VALUES (118, 18, 'Starry Night', 1889, 'Painting', 5000000)
```



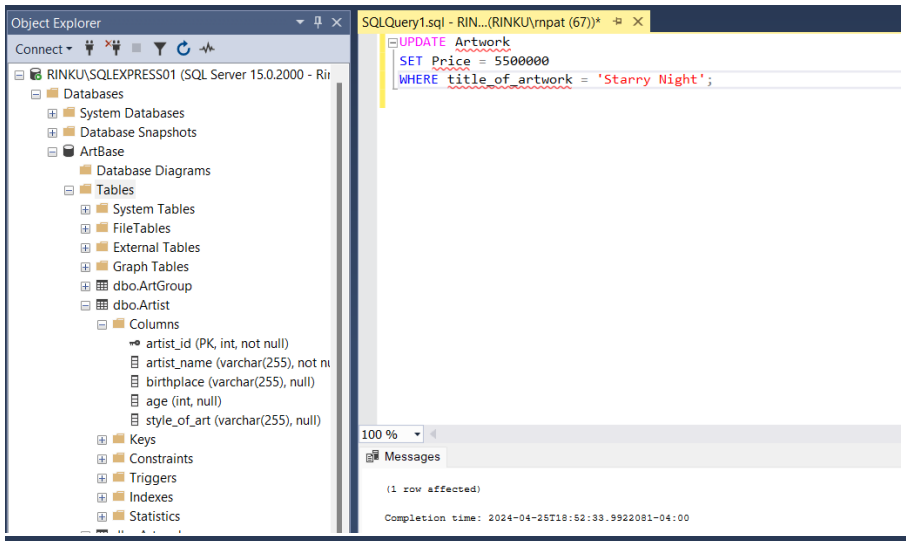


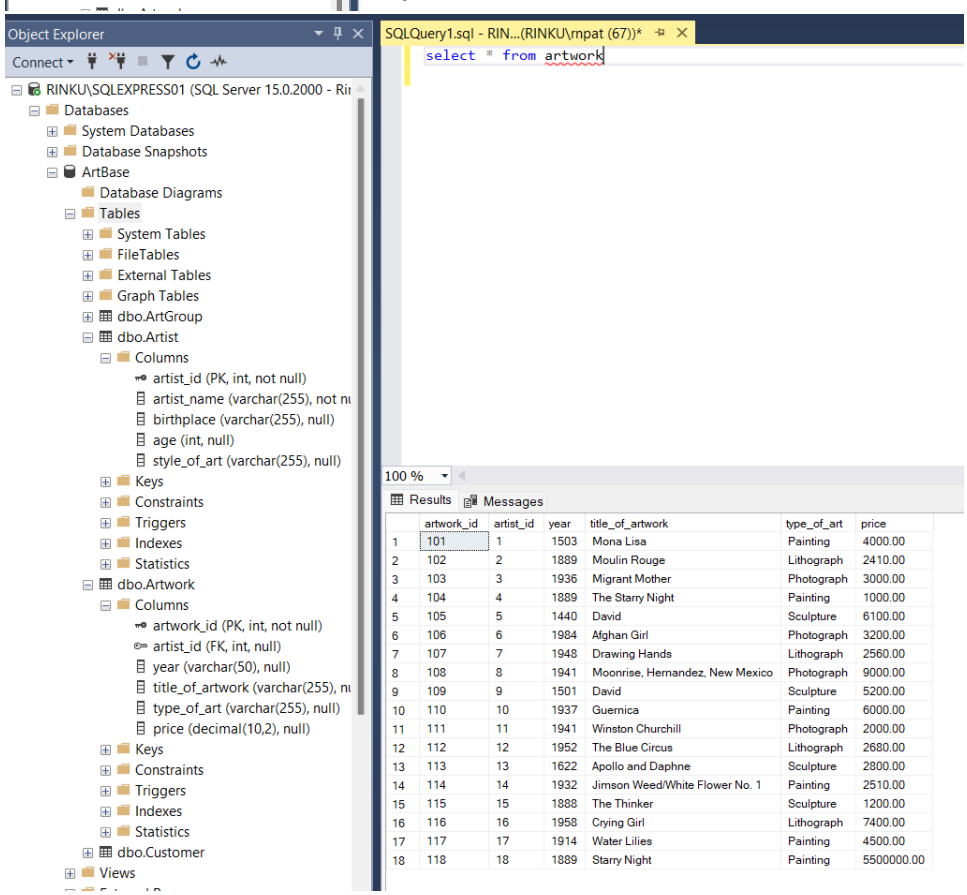
	artwork_id	artist_id	year	title_of_artwork	type_of_art	price
1	101	1	1503	Mona Lisa	Painting	4000.00
2	102	2	1889	Moulin Rouge	Lithograph	2410.00
3	103	3	1936	Migrant Mother	Photograph	3000.00
4	104	4	1889	The Starry Night	Painting	1000.00
5	105	5	1440	David	Sculpture	6100.00
6	106	6	1984	Afghan Girl	Photograph	3200.00
7	107	7	1948	Drawing Hands	Lithograph	2560.00
8	108	8	1941	Moonrise, Hernandez, New Mexico	Photograph	9000.00
9	109	9	1501	David	Sculpture	5200.00
10	110	10	1937	Guernica	Painting	6000.00
11	111	11	1941	Winston Churchill	Photograph	2000.00
12	112	12	1952	The Blue Circus	Lithograph	2680.00
13	113	13	1622	Apollo and Daphne	Sculpture	2800.00
14	114	14	1932	Jimson Weed/White Flower No. 1	Painting	2510.00
15	115	15	1888	The Thinker	Sculpture	1200.00
16	116	16	1958	Crying Girl	Lithograph	7400.00
17	117	17	1914	Water Lilies	Painting	4500.00
18	118	18	1889	Starry Night	Painting	5000000.00

3. **Update Queries:**
   - Update the price of an artwork:

Code:

```
UPDATE Artwork
SET Price = 5500000
WHERE title_of_artwork = 'Starry Night';
```





	artwork_id	artist_id	year	title_of_artwork	type_of_art	price
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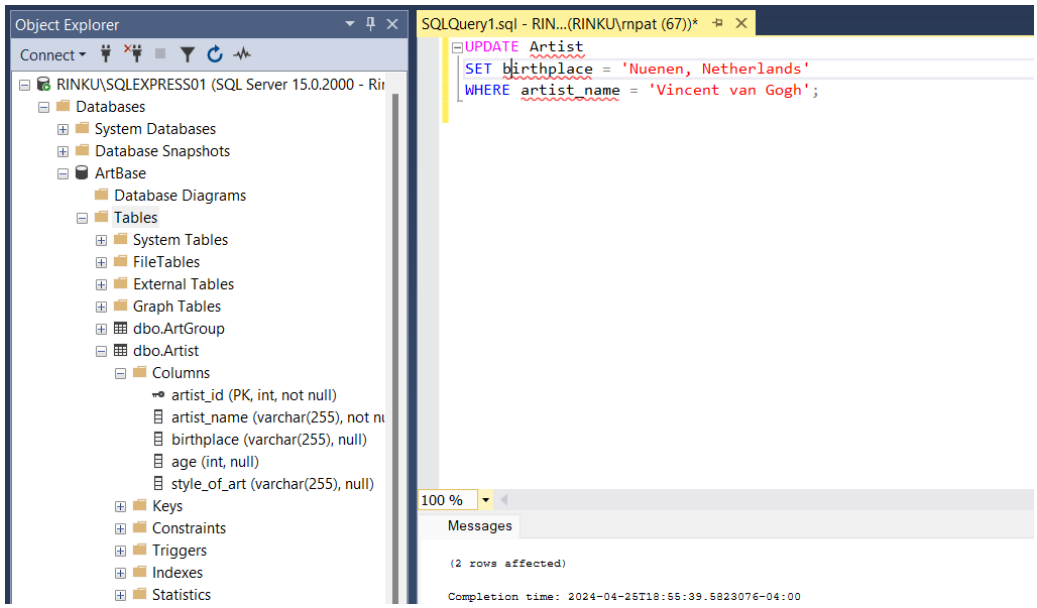
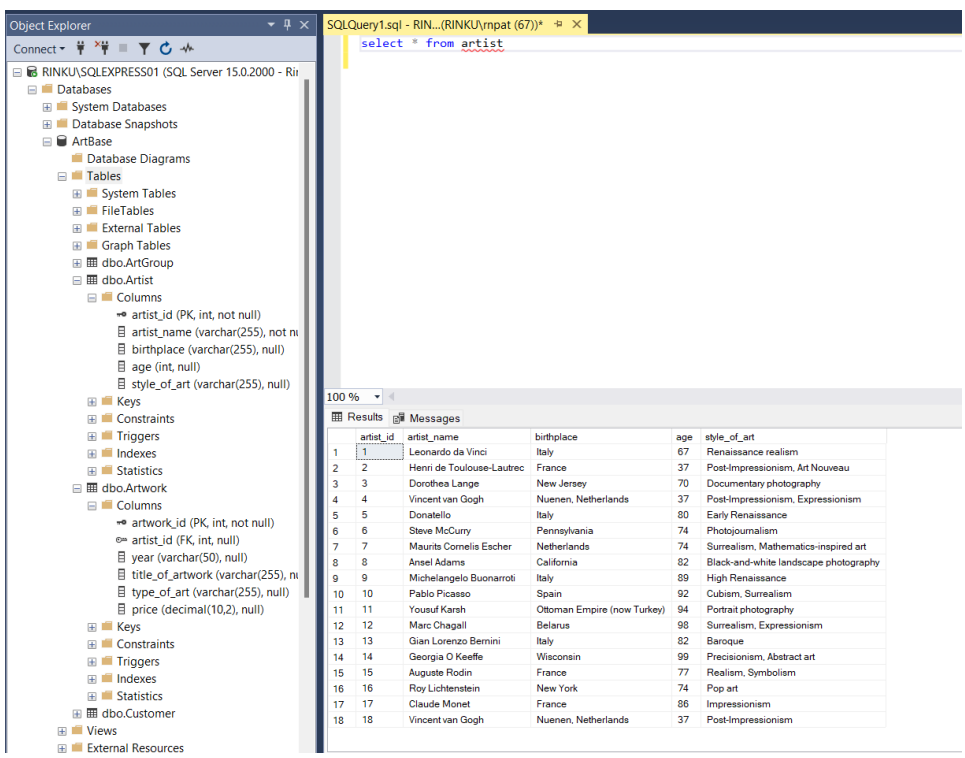
2	102	2	1889	Moulin Rouge	Lithograph	2410.00
3	103	3	1936	Migrant Mother	Photograph	3000.00
4	104	4	1889	The Starry Night	Painting	1000.00
5	105	5	1440	David	Sculpture	6100.00
6	106	6	1984	Afghan Girl	Photograph	3200.00
7	107	7	1948	Drawing Hands	Lithograph	2560.00
8	108	8	1941	Moonrise, Hernandez, New Mexico	Photograph	9000.00
9	109	9	1501	David	Sculpture	5200.00
10	110	10	1937	Guernica	Painting	6000.00
11	111	11	1941	Winston Churchill	Photograph	2000.00
12	112	12	1952	The Blue Circus	Lithograph	2680.00
13	113	13	1622	Apollo and Daphne	Sculpture	2800.00
14	114	14	1932	Jimson Weed/White Flower No. 1	Painting	2510.00
15	115	15	1888	The Thinker	Sculpture	1200.00
16	116	16	1958	Crying Girl	Lithograph	7400.00
17	117	17	1914	Water Lilies	Painting	4500.00
18	118	18	1889	Starry Night	Painting	5500000.00

- Update the birthplace of an artist:

Code:

```
UPDATE Artist
SET birthplace = 'Nuenen, Netherlands'
WHERE artist_name = 'Vincent van Gogh';
```



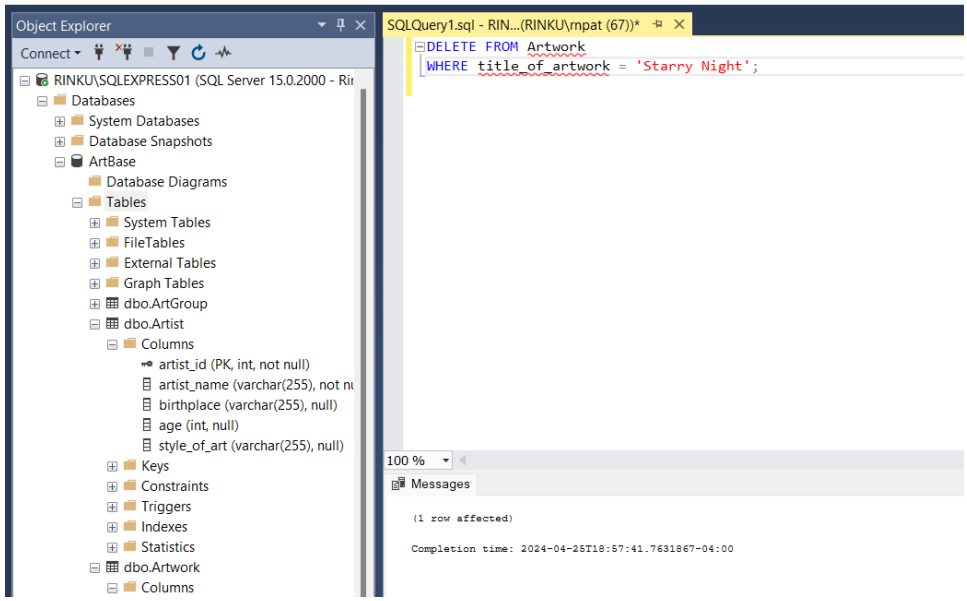


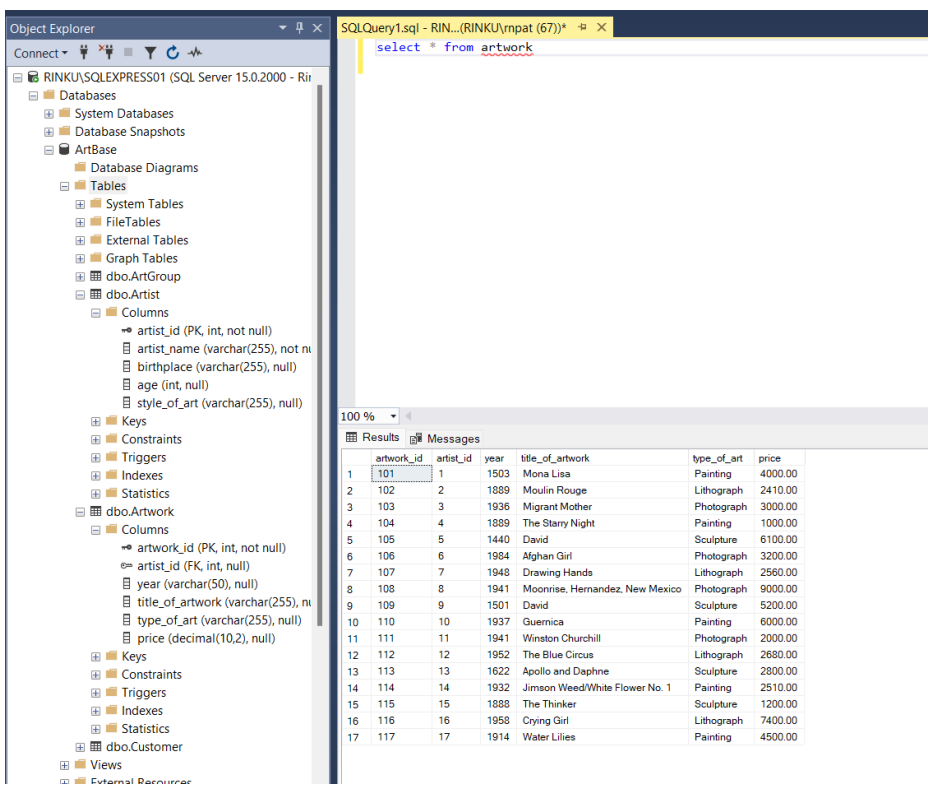
	artist_id	artist_name	birthplace	age	style_of_art
1	1	Leonardo da Vinci	Italy	67	Renaissance realism
2	2	Henri de Toulouse-Lautrec	France	37	Post-Impressionism, Art Nouveau
3	3	Dorothea Lange	New Jersey	70	Documentary photography
4	4	Vincent van Gogh	Nuenen, Netherlands	37	Post-Impressionism, Expressionism
5	5	Donatello	Italy	80	Early Renaissance
6	6	Steve McCurry	Pennsylvania	74	Photojournalism
7	7	Maurits Cornelis Escher	Netherlands	74	Surrealism, Mathematics-inspired art
8	8	Ansel Adams	California	82	Black-and-white landscape photography
9	9	Michelangelo Buonarroti	Italy	89	High Renaissance
10	10	Pablo Picasso	Spain	92	Cubism, Surrealism
11	11	Yousuf Karsh	Ottoman Empire (now Turkey)	94	Portrait photography
12	12	Marc Chagall	Belarus	98	Surrealism, Expressionism
13	13	Gian Lorenzo Bernini	Italy	82	Baroque
14	14	Georgia O Keeffe	Wisconsin	99	Precisionism, Abstract art
15	15	Auguste Rodin	France	77	Realism, Symbolism
16	16	Roy Lichtenstein	New York	74	Pop art
17	17	Claude Monet	France	86	Impressionism
18	18	Vincent van Gogh	Nuenen, Netherlands	37	Post-Impressionism

4. **Delete Queries:**
   - Delete an artwork:

   Code:

```
DELETE FROM Artwork
WHERE title_of_artwork = 'Starry Night';
```



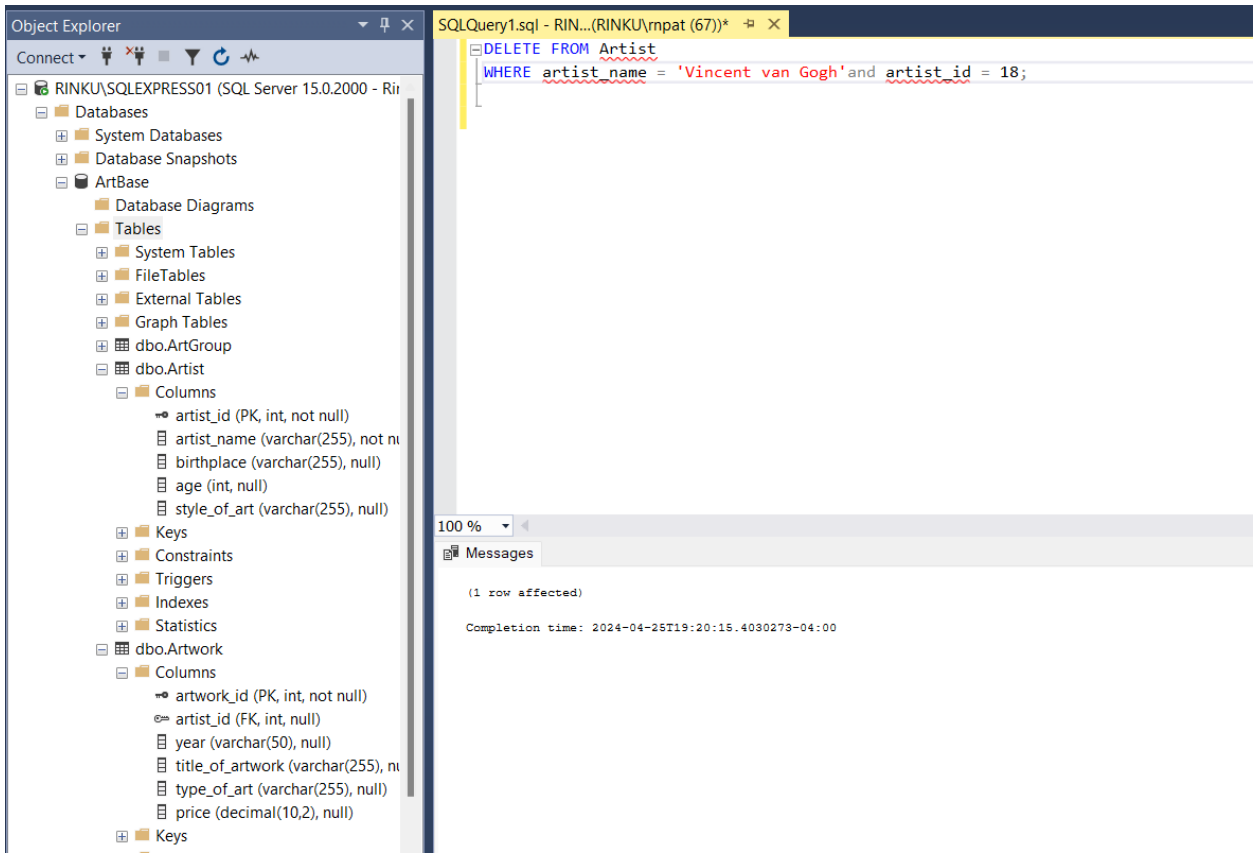


	artwork_id	artist_id	year	title_of_artwork	type_of_art	price
1	101	1	1503	Mona Lisa	Painting	4000.00
2	102	2	1889	Moulin Rouge	Lithograph	2410.00
3	103	3	1936	Migrant Mother	Photograph	3000.00
4	104	4	1889	The Starry Night	Painting	1000.00
5	105	5	1440	David	Sculpture	6100.00
6	106	6	1984	Afghan Girl	Photograph	3200.00
7	107	7	1948	Drawing Hands	Lithograph	2560.00
8	108	8	1941	Moonrise, Hernandez, New Mexico	Photograph	9000.00
9	109	9	1501	David	Sculpture	5200.00
10	110	10	1937	Guernica	Painting	6000.00
11	111	11	1941	Winston Churchill	Photograph	2000.00
12	112	12	1952	The Blue Circus	Lithograph	2680.00
13	113	13	1622	Apollo and Daphne	Sculpture	2800.00
14	114	14	1932	Jimson Weed/White Flower No. 1	Painting	2510.00
15	115	15	1888	The Thinker	Sculpture	1200.00
16	116	16	1958	Crying Girl	Lithograph	7400.00
17	117	17	1914	Water Lilies	Painting	4500.00

- Delete an artist:

Code:

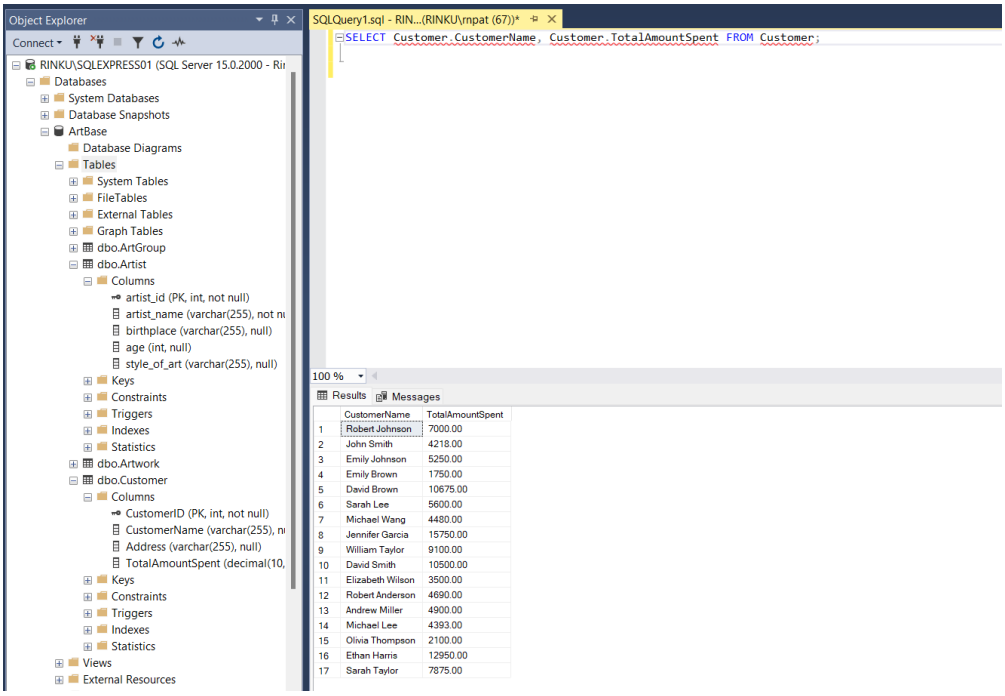
```
DELETE FROM Artist
WHERE artist_name = 'Vincent van Gogh'and artist_id = 18;
```



5. **Join Queries:**
   - Retrieve customer names and their total amount spent:

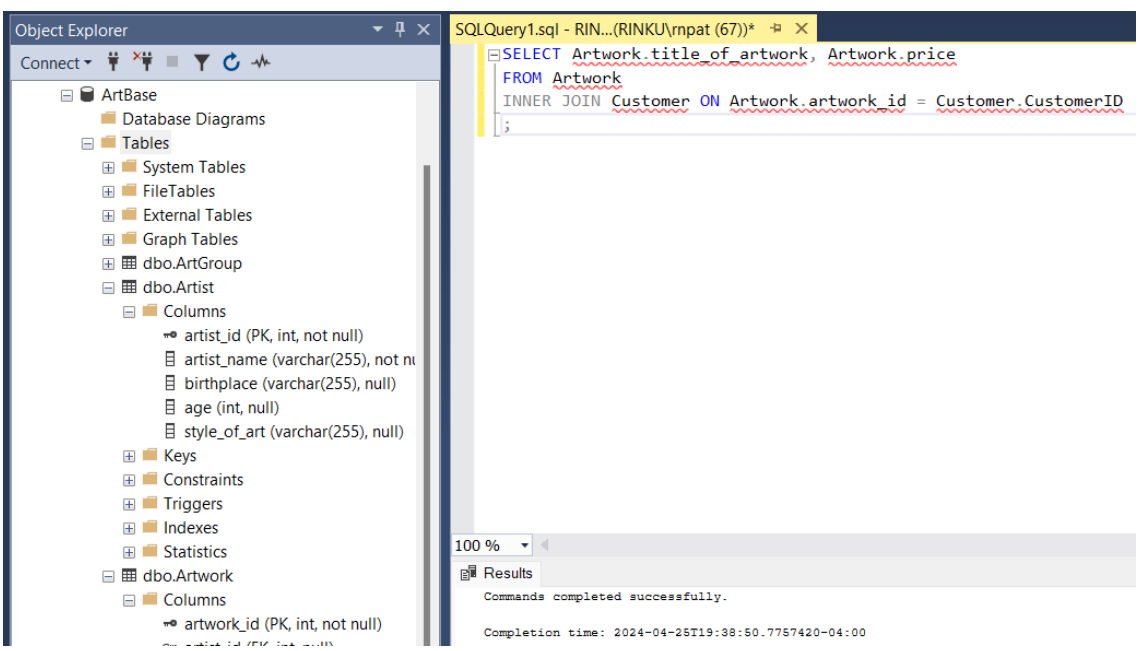
   Code:

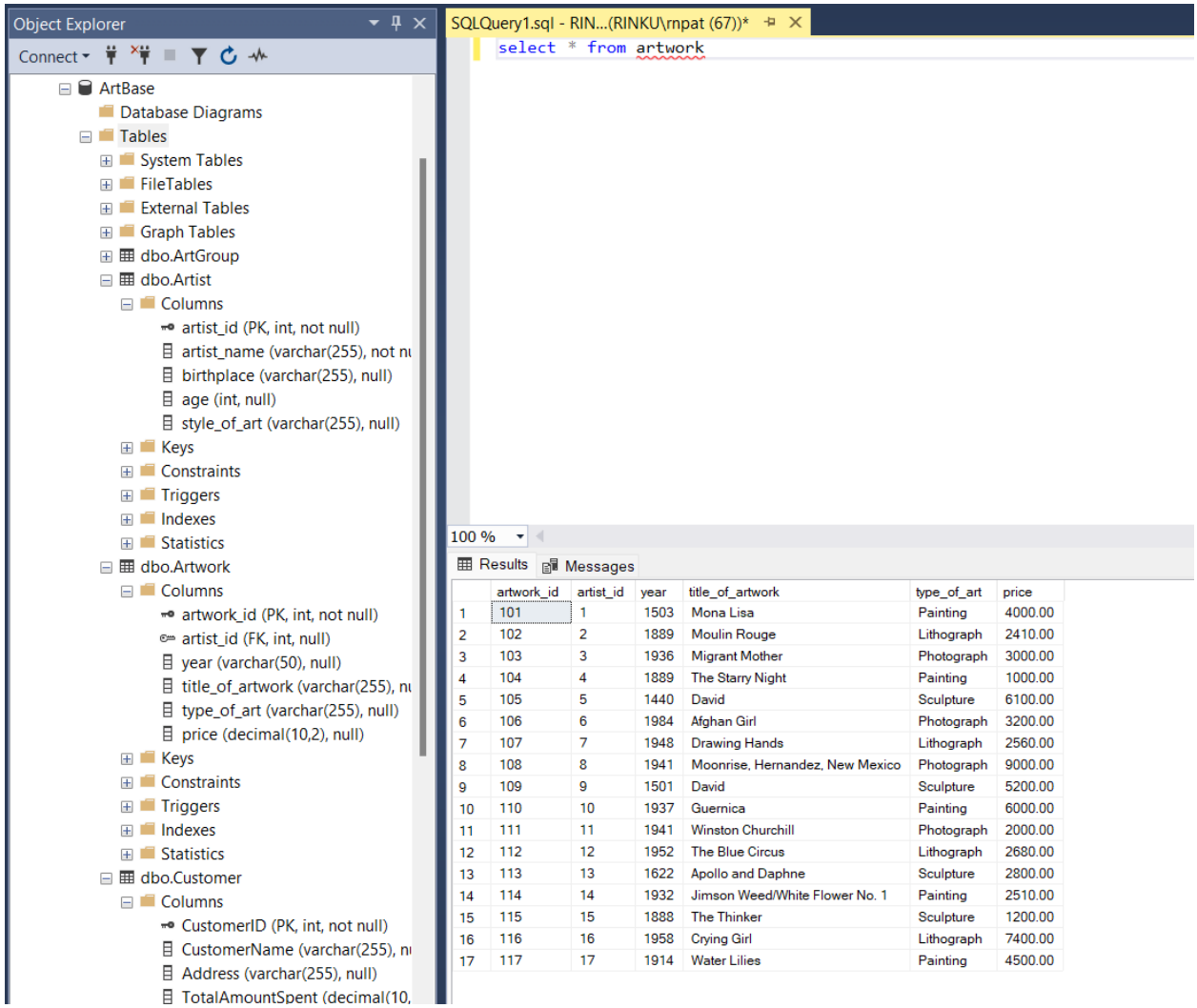
```
SELECT Customer.CustomerName, Customer.TotalAmountSpent FROM Customer;
```



- Retrieve artworks purchased by a specific customer:

sql
Copy code
SELECT Artwork.TitleOfArtwork, Artwork.Price FROM Artwork INNER JOIN Customer ON Artwork.CustomerID = Customer.CustomerID WHERE Customer.CustomerName = 'John Doe';

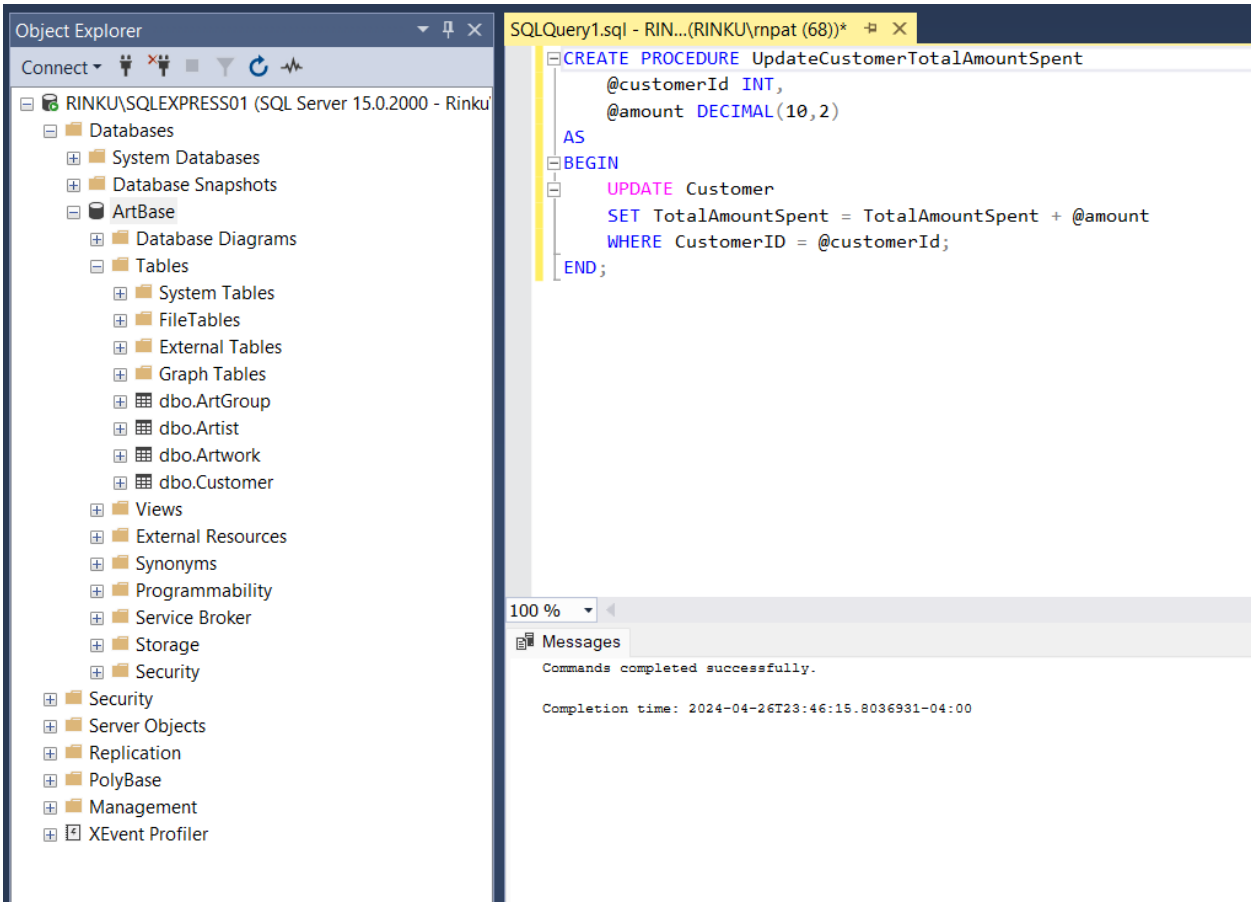




6. **Stored Procedures and Triggers:**
   - Example of a stored procedure:

   Code:

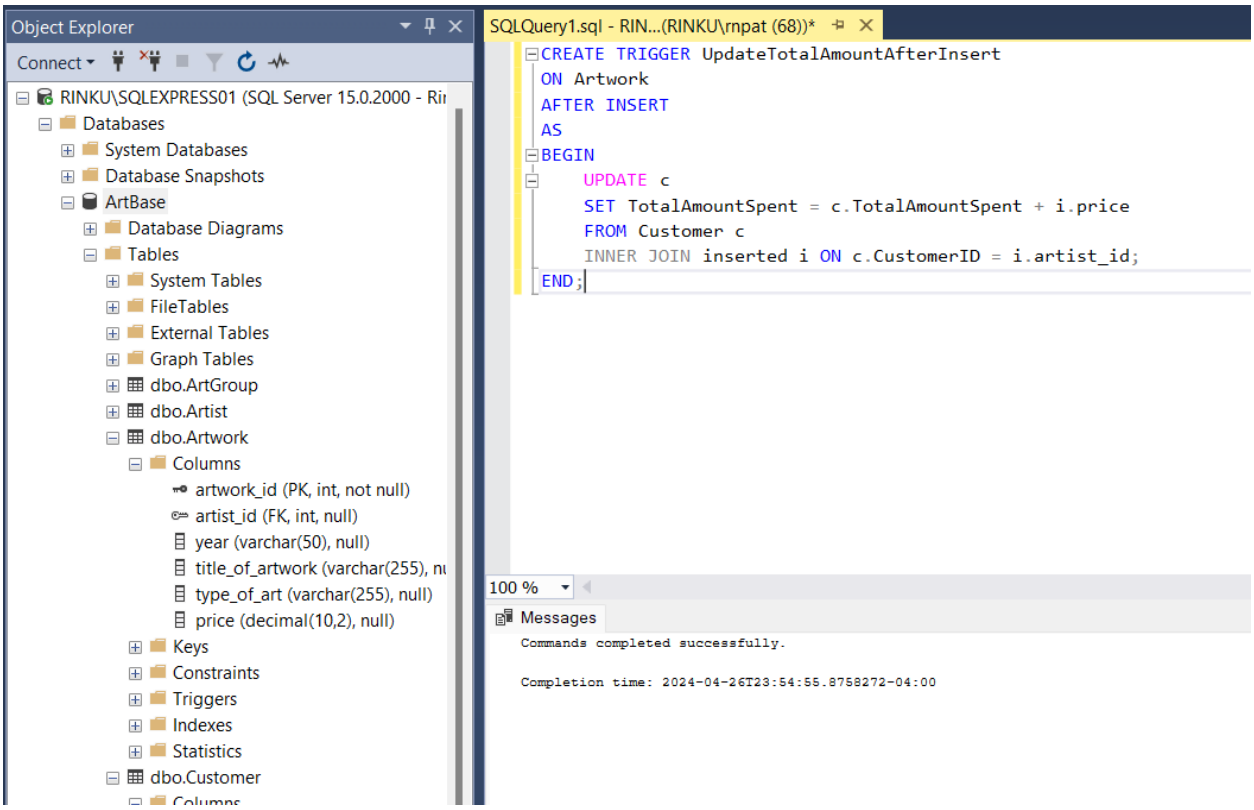
```
CREATE PROCEDURE UpdateCustomerTotalAmountSpent
    @customerId INT,
    @amount DECIMAL(10,2)
AS
BEGIN
    UPDATE Customer
    SET TotalAmountSpent = TotalAmountSpent + @amount
    WHERE CustomerID = @customerId;
       END;
```



- Example of a trigger (e.g., automatically update total amount spent when inserting a new purchase):

Code:

```
CREATE TRIGGER UpdateTotalAmountAfterInsert
ON Artwork
AFTER INSERT
AS
BEGIN
    UPDATE c
    SET TotalAmountSpent = c.TotalAmountSpent + i.price
    FROM Customer c
    INNER JOIN inserted i ON c.CustomerID = i.artist_id;
       END;
```

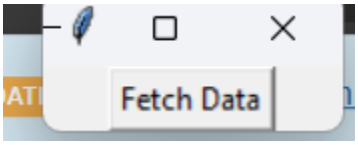


These SQL queries demonstrate the versatility and power of the ArtBase Database, enabling efficient data manipulation, retrieval, and maintenance. Stored procedures and triggers further enhance the functionality and automation of the database system, ensuring data integrity and consistency.

Application (optional) – You will earn extra credits (10 points) if you choose to develop an application such as .NET, Java, Python, and PHP with GUI to retrieve data from your database.

```
import tkinter as tk
from tkinter import ttk
import pyodbc

def fetch_data():
    try:
        # Replace 'Your_ServerName' and 'Your_DatabaseName' with the actual server and
database names
        server = 'RINKU\SQLEXPRESS01'
        database = 'ArtBase'

        # Establish connection to SQL Server
        conn = pyodbc.connect('DRIVER={SQL Server};SERVER=' + server +
                              ';DATABASE=' + database +
                              ';Trusted_Connection=yes;')  # Use Windows authentication
```

```
        cursor = conn.cursor()
        cursor.execute('SELECT * FROM Artwork')
        data = cursor.fetchall()

        # Display data in the Text widget
        root = tk.Tk()
        root.title('Artwork Data')
        text_area = tk.Text(root)
        text_area.pack()
        for row in data:
            text_area.insert(tk.END, ', '.join(map(str, row)) + '\n')  # Join row elements with commas
before inserting
        root.mainloop()
    except pyodbc.Error as e:
        print("An error occurred:", e)

root = tk.Tk()
root.title('Artwork Data Retriever')
fetch_button = tk.Button(root, text='Fetch Data', command=fetch_data)
fetch_button.pack()
root.mainloop()
```



Artwork Data

```
101, 1, 1503, Mona Lisa, Painting, 4000.00
102, 2, 1889, Moulin Rouge, Lithograph, 2410.00
103, 3, 1936, Migrant Mother, Photograph, 3000.00
104, 4, 1889, The Starry Night, Painting, 1000.00
105, 5, 1440, David, Sculpture, 6100.00
106, 6, 1984, Afghan Girl, Photograph, 3200.00
107, 7, 1948, Drawing Hands, Lithograph, 2560.00
108, 8, 1941, Moonrise, Hernandez, New Mexico, Photograph, 9000.00
109, 9, 1501, David, Sculpture, 5200.00
110, 10, 1937, Guernica, Painting, 6000.00
111, 11, 1941, Winston Churchill, Photograph, 2000.00
112, 12, 1952, The Blue Circus, Lithograph, 2680.00
113, 13, 1622, Apollo and Daphne, Sculpture, 2800.00
114, 14, 1932, Jimson Weed/White Flower No. 1, Painting, 2510.00
115, 15, 1888, The Thinker, Sculpture, 1200.00
116, 16, 1958, Crying Girl, Lithograph, 7400.00
117, 17, 1914, Water Lilies, Painting, 4500.00
```